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Sub-ns time transfer consistency: a direct comparison between GPS CV and T2L2

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Abstract

This paper presents a direct comparison between two satellite time transfer techniques: common-view (CV) of satellites from the global positioning system (GPS) constellation, and time transfer by laser link (T2L2) through the low orbiting satellite Jason-2. We describe briefly both techniques, together with two independent relative calibration campaigns of the links involving four European laboratories. Between the same remote time scale reference points, the mean values of the calibrated differences between GPS CV and T2L2 are below 240 ps, with standard deviations below 500 ps, mostly due to GPS CV. Almost all sample deviations from 0 ns are within the combined uncertainty estimates. Despite the relatively small number of common points obtained, due to the fact that T2L2 is weather dependent, these results provide an unprecedented sub-ns consistency between two independently calibrated microwave and optical satellite time transfer techniques.

Keywords: GPS common view, T2L2, calibration, time transfer uncertainty

(Some figures may appear in colour only in the online journal)

1. Introduction

The most common time transfer technique used for remote time scale comparisons is based on common-view (CV) of satellites from the global positioning system (GPS) constellation, one of the global navigation satellite systems (GNSS) currently in operation. In each station, a GPS receiver collects measurements of the differences between the local time scale and a satellite clock, which can be referred to GPS Time, the common satellite system time scale. Measurements can be based on P = code data broadcast on both GPS microwave carriers, allowing for the ionosphere-free P3 combination based on the software developed at the Royal Observatory of Belgium (ROB) [1] and made available by the Bureau International des Poids et Mesures (BIPM). Time transfer between

remote time scales is obtained by computing the CV differences between similar Common generic GNSS time transfer standard (CGGTTS) data recorded at the same epochs [2]. The data processing consists of removing outliers [3], in order to build a time difference mean value for each CGGTTS standard epoch. Another geodetic technique applied to time transfer between remote time scales, precise point positioning (PPP), as developed by National Resources Canada (NRCan) [4] among other groups, provides better short term stability because measurements are based on the phase of the carrier frequencies. But the carrier does not contain any identifiable time marker, and therefore the time transfer uncertainty remains limited by the detection capability of the time tag associated with the code. In addition, short term stability is not the issue when comparing the accuracy of different time

transfer technique. For this reason we disregard PPP in the following, any GPS time transfer uncertainty budget being limited by the same uncertainty on the determination of the hardware delays.

Time transfer by laser link (T2L2) is an optical two-way technique developed by the laboratory Géoazur of Observatoire de la Côte d'Azur (OCA), Valbonne, France, with the support of the French Space Agency CNES. The T2L2 principle is derived from classical satellite laser telemetry techniques, with dedicated on-board instrumentation able to time tag the laser pulses reaching the satellite [5]. Measurements for a given laser pulse consist of timing triplets based on the time of transmission from the ground, the time of arrival at the satellite, and the time of detection back on the ground of the photons reflected by the laser reflector array of the satellite [6]. The time transfer between two stations is obtained from the combination of such triplets, after computation of a fit for each satellite path producing one normal point [7] of the time scale difference. An improvement of at least one order of magnitude in time transfer uncertainty over the existing microwave techniques is expected. In addition, on the basis of this T2L2 time transfer uncertainty, a simple averaged frequency between two T2L2 points over a given time interval would provide the capability to compare today's most accurate frequency standards. T2L2, based on a ground network of laser ranging stations and on a dedicated space segment, is currently in its operational phase since the launch in 2008 of the dedicated payload hosted by the Jason-2 satellite [6, 8]. The years 2012–2013 were especially devoted to T2L2 accuracy estimation [9–11], T2L2 long term time stability, fundamental physics and comparison between T2L2 and GPS CV [12]. One of the goals of the T2L2 experiment, which by its nature can only provide relatively sparse measurements, is in fact the one-off calibration of other time transfer techniques. Part of this T2L2 campaign took place between OCA, NERC Space Geodetic Facility (SGF) at Herstmonceux Castle, UK, and Geodetic Observatory Wettzell (GOW), Germany. In all stations, the reference clocks for time measurements were active H-maser standards.

During Autumn 2013, the OCA traveling laser station called FTLRS [13] was installed at Observatoire de Paris (OP), Paris, France, for tests on frequency transfer. FTLRS was then connected to the active H-maser used as local oscillator for the LNE-SYRTE Primary Frequency Standards, LNE-SYRTE being the OP laboratory designated by the French National Metrology Institute (NMI) Laboratoire National de Métrologie et d'Essais (LNE) for Time and Frequency Metrology activities. This enabled OP to participate in T2L2 sessions as a laser ranging station. A direct comparison between GPS CV and T2L2 among the participating stations could then be determined. For that purpose, a double campaign for the characterization of the link offsets was organized by LNE-SYRTE for the GPS CV, presented in section 2, and by OCA for T2L2, presented in section 3. The H-maser s are free running in OP, SGF and GOW. But is steered by a micro phase stepper in OCA, in order to keep the local time scale close to coordinated universal time (UTC). In OCA, OP and SGF, a 10 MHz signal and a 1 pulse per second (PPS) signal,

used as reference time scale, were locally distributed to both GPS and T2L2 equipment by common distribution amplifiers. Unfortunately, the laser station and the GPS station in GOW were not directly connected to the same H-maser, and building of an accurate relationship between both proved not to be an easy task. Therefore, GOW was eventually not included in the GPS hardware delay relative offset calibration campaign, and does not appear in the analysis of the direct comparison between these independently calibrated techniques presented in section 4.

2. GPS link calibration

Initially, the idea was to carry out a classic relative measurement campaign with a multichannel geodetic GPS receiver travelling unit, in order to characterize the differential hardware delays between a reference receiver located in OP and remote receivers in OCA and SGF [14]. But this campaign was essentially aiming at a direct comparison with T2L2, and the station network was limited to three locations only. Therefore, a dedicated travelling system was put together in order to provide a direct calibration of the links between given reference points of the remote time scales, in addition to the characterization of the differential hardware delays. The campaign discussed here took place from September to November 2013. In each station, the local time scale reference point for link calibration measurements is taken as the 1.0V level of the rising edge of a specific output of the 1 PPS distribution amplifier. The OP travelling equipment consisted of two geodetic receivers of different types and manufacturers and one single Choke-Ring antenna, along with an ensemble of signal generation and distribution, plus a time interval counter. The results were corrective offsets between remote time scales for the links OCA–OP, OP–SGF and OCA–SGF, together with the related uncertainties. The whole campaign and its analysis are related in [15].

During this campaign, the involved geodetic GPS receivers were as follows. In OP, an Ashtech Z12-T of first generation called OPMT served as reference receiver for all computations. It is the main unit of the International GNSS Service (IGS) station implemented in OP. In OCA, a DICOM GTR50 called OC03 is implemented nearby the laser station. In SGF, a Septentrio PolaRx3 is the main unit of the IGS station HERS. In addition the OP travelling units were an Ashtech Z12-T of second generation called OPM4, and a Septentrio PolaRx4 called OPM7. The NRCan PPP processing technique was used for the accurate computation of the antenna coordinates at the time of the measurement campaign.

The expanded uncertainties of the GPS CV links corrected by the resulting offsets were estimated at 2.1 ns for OCA–OP and for OCA–SGF, and 1.5 ns for OP–SGF ($k = 2$). The uncertainty on the OP–SGF link is limited by the deviation from closure between the travelling equipment and the reference receiver in OP, which exhibited a typical value of about 0.7 ns. But we noticed a large deviation of more than 1 ns on GPS CV between the two OP travelling receivers OPM4 and OPM7 when located in OCA, which was unexpected. We

currently have only limited clues as to the cause of this discrepancy. On the one hand, we know that Septentrio PolaRx receivers are sensitive to the power level of the 10 MHz IN signal, which is used to generate internally the reference for tracking GNSS signals. Even with local reference delay measurements based on PolaRx 1 PPS OUT signal, it is possible that a small offset remains present in the GPS data when compared to other locations. On the other hand, the second generation of Ashtech Z12-T, which is rack-mounted contrary to the first generation based on a field geodetic receiver, appeared slightly more sensitive to temperature changes even when located inside a temperature controlled room. We detected a small diurnal term in the Allan time deviation (TDEV) between OPMT and another rack-mounted Ashtech Z12-T called OPM2 [16], despite the fact that both units were implemented side by side in a temperature controlled room and that both were connected to the same 3S navigation temperature stabilized antenna TSA-100. We have not made any further investigations on the subject yet, but it might be that both effects were present together, increasing the deviation between the receivers when located in OCA, but with small consequences on the average between both, which is eventually used for the link calibration computation to OCA time scale reference point. In this way, the results obtained by direct comparison to T2L2 do not exhibit any unexpected deviation, as can be seen in section 4 below. The deviation between OPM4 and OPM7 was nevertheless considered as the best estimator of the travelling equipment stability during the campaign for OCA link offsets, and this resulted in larger uncertainties on OCA–OP and OCA–SGF compared to OP–SGF.

3. T2L2 link calibration

A main characteristic of the T2L2 system is that it relies on the existing network of satellite laser ranging (SLR) stations on ground. However, laser ranging timing needs, especially in terms of time accuracy, are far less stringent than T2L2 objectives require. Thus during the last few years, specific effort has been made in the framework of the T2L2 project to develop new technologies to improve time distribution and calibration of laser stations. Several developments have led to the definition of very accurate measurement devices. The objectives were: (i) to allow for the time calibration of any SLR station; this quantity, noted C_{Cal} hereafter, corresponds to the total propagation delay between the SLR station (cross-axes of the telescope) and the reference clock (PPS distribution unit reference point), (ii) to improve the timing accuracy of SLR measurements, (iii) to design a dedicated travelling equipment package. Some of the particular challenges that have been studied are:

- (i) analysis of laser pulse width in the range of 20 ps–30 ps,
- (ii) stability of event timers at the sub-ps level, [17],
- (iii) quality of time and frequency distribution based on an ultra-stable 1 PPS signal generator, [18],
- (iv) measurement of the optical to electrical detection delay at detector level.

The implementation of the equipment involved in the measurements for both techniques is quite different from one SLR station to the other. In OCA, the distance between the time scale reference point of the station and the location of event timers is approximately 70 m [11]. In OP, the mobile laser station FTLRS was implemented on the roof of the building, not far from the GPS antennas. Dedicated cables with high protection against thermal gradients were installed between the roof and the LNE-SYRTE laboratory in the building where the reference point of the clock distribution unit is located. At both places, OP and OCA, a dedicated technique was developed to ensure continuous monitoring at ps level of the time delays between the different equipment for the duration of the campaign [18]. The time calibration at OCA station took place in October 2013, whereas the calibration of FTLRS delays while installed at OP was done twice during the campaign, in September and in October 2013. At SGF, the distances between collocated equipment, clocks, laser station and GPS receiver, are much shorter. Therefore, monitoring of the variation of the time delay in the cables was not required. The SGF laser station was calibrated in October 2013, in parallel with the GPS calibration. As already stated in the Introduction, in GOW some difficulties were encountered in the calibration process due to the large distances between involved laboratories and to the complexity of precisely identifying the time and frequency. As a consequence, we made a decision to save the acquired data but exclude the results from the calibration process.

The travelling equipment used for the T2L2 laser station calibration is based on a Sigma Time STX ps event timer and two photo diodes. One is a high speed photo detector, able to measure the absolute time delay between the rising edge of the laser pulse at the invariant reference point of the telescope and the electrical reference coming from the designated output of the PPS distribution unit of the time and frequency laboratory. The second is a single-photon avalanche diode (SPAD) detector working in Geiger mode in order to measure the laser pulse width. The measurements are achieved using an optical module placed at the output of the telescope and connected to the photo-diodes through a set of calibrated optical fibres. The hardware technique developed to obtain the calibration C_{Cal} is based on a two step measurement for a given station, as: $C_{\text{Cal}} = \delta_T + \delta_{\text{PRG}}$. Firstly, δ_T is the difference between the absolute epoch given by STX and the epoch of the laser pulse given by the local event timer of the laser station. Secondly, δ_{PRG} represents the sum of the propagation delays of all the different parts of the measurement chain, from the cross-axes of the telescope to the PPS distribution unit reference point.

Table 1 shows the calibration values C_{Cal} measured at each SLR station using the same travelling equipment. The definition of the propagation term through the telescope to the PPS distribution unit is given in figure 4 in [10] for the main characteristics of the T2L2 calibration station. The calibration uncertainty is deduced from the quadratic sum of many terms depending on the instrumentation involved in the measurement; the details of these terms can be found in table 8 in [10]. Note that the difference between the September and October

Table 1. T2L2 calibration values (in ps) of the SLR system on each geodetic site involved during the campaign.

	Calibration C_{Cal} (ps)	Uncertainty (ps)	Date
OCA	236 601.46	33.9	
SGF	195 837.46	34.9	2013-10-22
OP	203 343.54	33.5	2013-09-17
(FTLRS)			
	203 366.46	34.2	2013-10-24
GOW	10 575.2	50.0	2013-02-26

delay calibrations of FTLRS while installed at OP is below 23 ps, in line with the estimated uncertainties.

The propagation of a high precision 1 PPS reference signal requires an ultra stable distribution between the reference clock and the SLR station. Considering a thermal gradient in the range of 20 K, the total delay in the cables can vary up to 200 ps which is significant for the campaign target. As a result of technological research and development, an ultra stable time signal generator SigmaTime STS201 was designed, the repeatability error of which is better than 1 ps rms for the 1 PPS distribution [18]. During the campaign, one STS instrument was installed in OCA and another in OP, in order to continuously measure the propagation delays ($\epsilon_S(t)$, at each site S) in the ground cables. A series of data have been acquired in both sites, with a 20 s sampling period. As can be seen in figure 1 there is a peak to peak difference between OCA and OP of about a factor 3, but both data sets follow a similar longer term fluctuation, almost certainly a weather dependent behaviour despite the distance between the stations. There is heating by sunlight during the day and the air gradually cools overnight. The larger variation observed in OP data seems primarily due to the fact that a part of the cable run was outside on the roof exposed to this temperature effect, indicating that the protection around these cables was not sufficient for measurements at ps level. The peak to peak variation in OCA does not exceed 60 ps over 60 d. These measurements were injected in the data processing in order to slightly correct each ground-to-space link, and this is done just before the ground-to-ground time transfer computation [7].

4. Direct comparison between both techniques

During the comparison period, all GPS stations operated almost continuously. By using the offsets between remote time scale reference points obtained from the GPS calibration campaign, the time differences between the time scales have been computed by GPS CV for the three links OCA–OP, OCA–SGF and OP–SGF. The classical GPS CV are 16 min sampled. But we also computed a filtered GPS time transfer by a moving average over 13 consecutive CGGTTS CV epochs. This provides 16 min sampled 3.5 h averaged points exhibiting a lower noise, the averaging period having been chosen in order to avoid smoothing out artificially any real physical effects included in the data.

In contrast to this continuous GPS CV data set, the T2L2 normal points provide the time differences between two remote time scales at epochs depending on the satellite path and available laser ranging triplets [8].

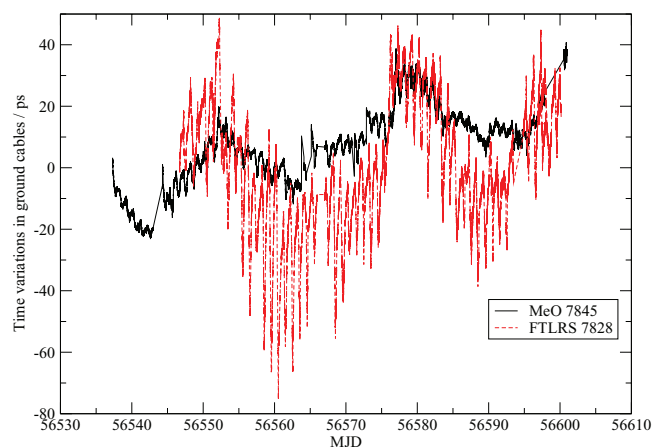


Figure 1. Monitoring of the time variations (in ps) of signals in ground cables from the SLR station to the local clock distribution reference point at OP (in red: FTLRS 7828) and in OCA (in black: MeO 7845), based on Sigma Time STS time signal generators. The correlation between plots is almost certainly due to similar weather conditions despite the distance between both stations.

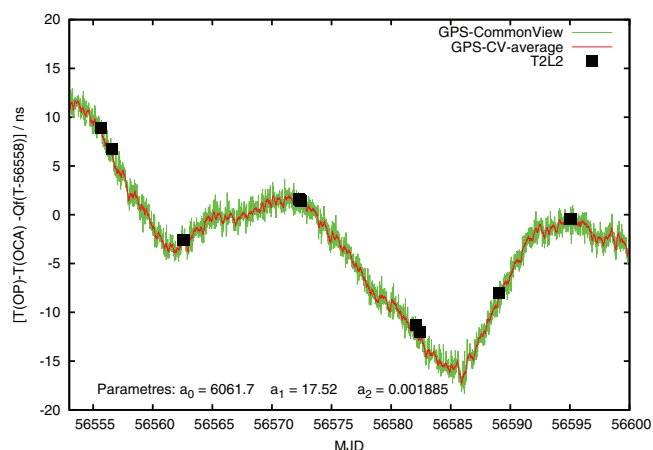


Figure 2. Time differences between time scale reference points in OCA and in OP with a quadratic fit removed: green continuous noisy line is for GPS CV, red smoothed continuous line is for GPS CV filtered by a moving average over 13 samples, and blue squares are for T2L2.

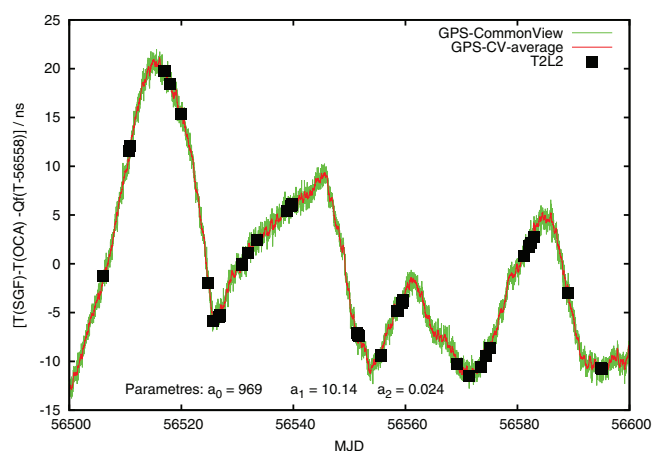


Figure 3. Time differences between time scale reference points in OCA and in SGF with a quadratic fit removed: green continuous noisy line is for GPS CV, red smoothed continuous line is for GPS CV filtered by a moving average over 13 samples, and blue squares are for T2L2.

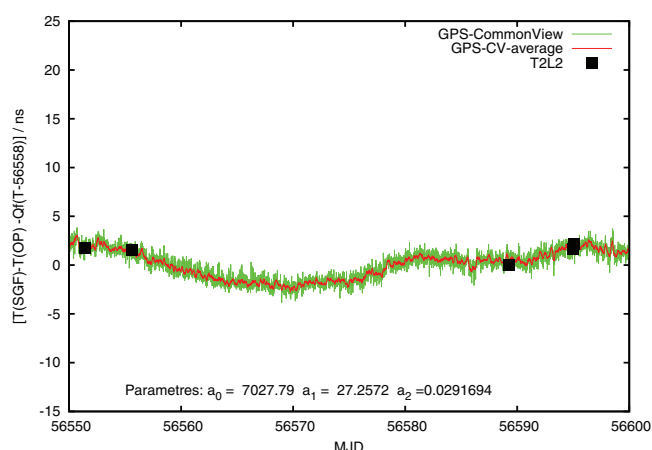


Figure 4. Time differences between time scale reference points in OP and in SGF with a quadratic fit removed: green continuous noisy line is for GPS CV, red smoothed continuous line is for GPS CV filtered by a moving average over 13 samples, and blue squares are for T2L2.

Table 2. Mean values of the time differences between GPS CV and T2L2.

Link	Number of points	Mean values (ns)	Std deviations (ns)		Combined uncertainty (ns)
			Raw GPS CV	Filtered	
OCA–OP	12	0.24	0.48	0.25	1.1
OCA–SGF	42	0.09	0.49	0.37	1.1
OP–SGF	5	0.10	0.32	0.32	0.8

For each link, we have plotted both GPS CV time transfer results, together with the T2L2 results. For clarity of the figures, an additional quadratic fit was also removed, in order to take into account the H-maser differential drifts: the parameters a_0 , a_1 and a_2 of the fit $a_0 + a_1t + a_2t^2$ are provided inside each figure. Figure 2 shows such time scale differences derived from both techniques on the link OCA–OP: the higher jitter continuous green line shows the GPS CV 16 min sampled time transfer, the smoothed continuous red line shows the additionally 3.5 h averaged GPS CV time transfer, and the blue squares are the T2L2 time differences. Similarly, figures 3 and 4 show the time scale differences on the links OCA–SGF and OP–SGF, respectively. Note that in the plots including OCA, the shape of the curves is due to the manual steering of the H-maser in order to keep the local time scale close to UTC. One can see that there are only a relatively small number of T2L2 points on the plots. This is mostly due to the fact that T2L2 is weather dependent, but also to some technical issues which limited the measurements for some links. The largest number of common points were obtained for the OCA–SGF link compared to those including OP, this simply related to the relative volume of laser ranging data obtained by the 3 participating stations. But we also see that the discrepancy between the independently calibrated techniques remains remarkably small over the whole campaign period for all links.

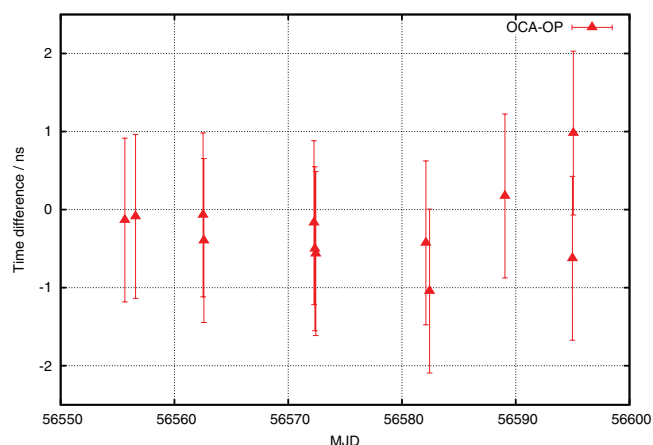


Figure 5. Differences between time transfer obtained from GPS CV and T2L2 on the OCA–OP link, the error bars showing the estimated combined uncertainty of about 1.1 ns.

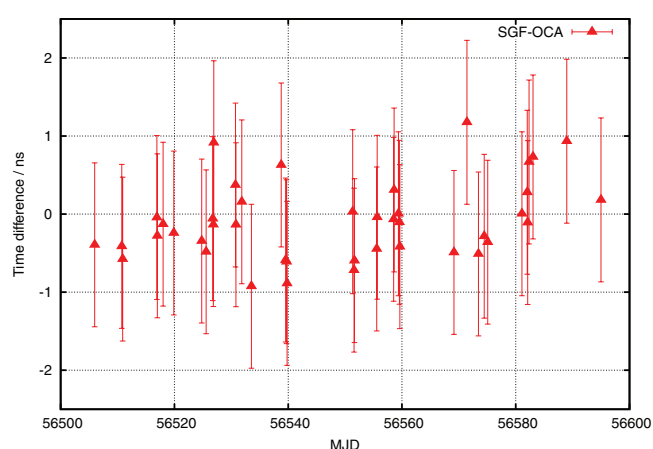


Figure 6. Differences between time transfer obtained from GPS CV and T2L2 on the OCA–SGF link, the error bars showing the estimated combined uncertainty of about 1.1 ns.

GPS CV analysis produces 16 min sampled time scale differences which are not necessarily synchronous with any T2L2 normal point epochs. For the computation of the differences between techniques, two adjacent raw GPS CV points have been interpolated to produce one GPS CV result synchronous with the corresponding T2L2 normal point epoch. Figures 5–7 show the deviations between techniques for the links OCA–OP, OCA–SGF and OP–SGF respectively. The combined uncertainties were computed by building a simple quadratic sum between the GPS CV link uncertainties given in [15], and the ground to ground T2L2 uncertainty given in [10]. We obtain 1.1 ns for OCA–OP and OCA–SGF, and 0.8 ns for OP–SGF. When properly calibrated, the two techniques are measuring the same time deviations between remote time scale reference points, hence the difference is expected to be close to zero. What the results indicate is the remarkable consistency between GPS CV and T2L2 techniques. Almost all time differences sampled are close to 0 ns within the combined uncertainties. In addition, we do not detect any significant trend nor periodic term, even when considering the OCA–SGF link with its greater quantity of data.

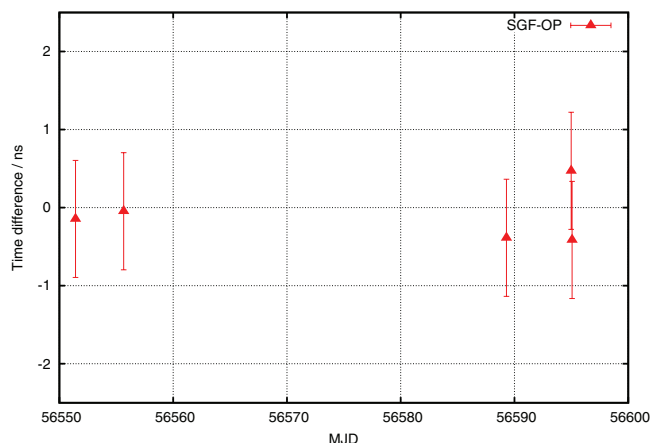


Figure 7. Differences between time transfer obtained from GPS CV and T2L2 on the OP–SGF link, the error bars showing the estimated combined uncertainty of about .8 ns.

The mean values of the three data sets are reported in table 2, together with the standard deviations and the combined uncertainties. We consider here the standard deviations obtained from the 16 min sampled GPS CV and those obtained after the additional smoothing on 13 consecutive samples. The average differences obtained between GPS CV and T2L2 are never greater than 240 ps. Despite the relatively small number of common points obtained, this clearly demonstrates a sub-ns consistency between the two independently calibrated techniques. Additionally, all deviations of the mean values from 0 ns are largely within the combined uncertainties. When considering the link OCA–SGF only, where the number of available points is significant enough to build a consistent mean value of the differences together with the related standard deviation, we can see that the moving average method of processing the GPS data reduces the noise significantly. But, despite this improvement and as expected, the GPS CV are setting not only the noise level but also the uncertainty for such data.

5. Conclusion

For the two time transfer techniques GPS CV and T2L2, two independent relative calibration campaigns were undertaken in Autumn 2013 between the three European stations OCA, OP and SGF. By using the corrective offsets obtained for each link between remote time scale reference points, common to both microwave and optical satellite techniques, a direct comparison provided an unprecedented consistency at a sub-ns level. The mean values of the time differences between GPS CV and T2L2 are no greater than 240 ps, with a standard deviation below 500 ps, mostly due to GPS CV. The resulting average differences are clearly negligible with respect to the combined uncertainties of both techniques, which are between 0.8 ns to 1.1 ns depending on the link considered. Although the estimated uncertainty of T2L2 is much lower than the estimated uncertainty of GPS CV, the good agreement between the two techniques confirms that the independent link calibrations have been carried out properly and that the uncertainty

budgets can be trusted, even when estimated conservatively. This result provides a full validation of the GPS CV link uncertainty budget computations, the expanded uncertainties of which were estimated between 1.5 ns to 2.1 ns ($k = 2$) on the considered links [15]. The result is also in line with the claimed T2L2 performance, the expanded uncertainty of which was estimated at 138 ps ($k = 2$) for ground to ground time transfer [10]. It is also a demonstration of the capability of T2L2 to undertake sub-ns calibration of other time transfer techniques.

The decommissioning of the space segment of T2L2, which was considered in 2014–2015, has been postponed, and the operation extended at least until the end of 2016. It should also be noted that a similar instrument called European laser timing (ELT) [19] will be hosted on-board the International Space Station (ISS) inside the atomic clock ensemble in space (ACES) module [20]. The launch of the ACES module is currently planned for mid 2017. As we believe has been demonstrated by the results described here, optical satellite techniques like T2L2 should continue to be considered as possible candidates for the validation of long distance time transfer through optical fibre links. Provided that the required portable laser stations can be made available in the reference sites and suitable flight instruments are implemented on-board new satellites in the future.

6. Disclaimer

Product names and model numbers of the equipment are included for reference only. No endorsement or criticism is implied.

Acknowledgments

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